A Reproducible Forecasting Protocol for African Central Banks

Walk-Forward Validation, Diebold–Mariano and Model Confidence Set Tests,

Optimal Bates–Granger Combination.

An Application to Côte d'Ivoire's CPI (1997-2026)

Boni MONNEY Jean

Direction Générale de l'Économie (DGE)

Ministry of Economy, Finance and Budget — Republic of Côte d'Ivoire

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Abstract

This paper proposes a reproducible inflation forecasting protocol designed for the operational constraints of African central banks, and applies it to the harmonised consumer price index (IHPC) of Côte d'Ivoire over 350 monthly observations spanning January 1997 to February 2026. The protocol rests on four methodological components. The first is walk-forward (rolling origin) validation with 18 forecast origins and four evaluation horizons ($h \in \{1, 3, 6, 12\}$), which replaces the fixed train-test split still widely used in African applied work. The second is systematic evaluation of forecasts on six metrics (MAE, RMSE, MASE, MAPE, SMAPE, Theil's U), which makes it possible to verify that the conclusions do not depend on the choice of loss function. The third is formal statistical inference, through the Diebold–Mariano test with the Harvey–Leybourne–Newbold small-sample correction and the Model Confidence Set procedure of Hansen et al. (2011), implemented with circular block bootstrap and 5,000 replications. The fourth is the optimal Bates–Granger (1969) combination with covariance-adjusted weights. Five models are compared: ARIMA with automatic AIC order selection, Holt-Winters triple exponential smoothing, random walk with drift, long short-term memory (LSTM) and the neural hierarchical interpolation architecture (NHITS). The application yields three results. First, NHITS achieves a MASE of 0.94 at $h = 1$, becoming the only model that beats the in-sample random walk benchmark. Second, Holt-Winters is the unique element of the Model Confidence Set at $h = 12$ and $\alpha = 25\%$, making it the robust choice for annual forecasting. Third, the Bates–Granger combination of NHITS and Holt-Winters produces a measurable 4.84% gain at the quarterly horizon and yields a twelve-month inflation forecast of +1.52%, comfortably inside the BCEAO's 1–3% target band. The complete code is released under the MIT license to facilitate adoption of the protocol by other African economies.

Keywords: inflation forecasting; ARIMA; Holt-Winters; LSTM; NHITS; walk-forward validation; Diebold–Mariano test; Model Confidence Set; Bates–Granger combination; African central banks; Côte d'Ivoire; WAEMU.

JEL Classification: C22 (Time-Series Models), C45 (Neural Networks), C53 (Forecasting Methods), E31 (Inflation), E37 (Forecasting and Simulation).

1. Introduction

Inflation forecasting occupies a central place in the conduct of monetary policy and in fiscal programming. For African central banks, and particularly those operating within monetary unions like WAEMU or CEMAC, reliable forecasts are indispensable to meeting convergence criteria and to the credibility of policy announcements. The Central Bank of West African States (BCEAO) targets a Union-wide inflation band of 1 to 3% per year, and persistent deviations from this band — as observed during the 2022 inflation surge tied to the Russia–Ukraine conflict and post-pandemic supply disruptions — quickly translate into credibility costs and complicate multi-year programming.

Forecasting practice in African economic administrations remains marked, however, by heterogeneous evaluation protocols, which make results difficult to compare across studies and across countries. Three methodological limitations are recurrent. First, forecast evaluation often rests on a single fixed train-test split, which makes conclusions sensitive to the partition date and offers little assurance of generalisation. Second, comparisons between models rely on point estimates of error metrics, without formal statistical tests that would distinguish a real difference from sampling variation. Third, forecast combination, whose interest has long been established in the econometric literature (Bates and Granger, 1969; Clemen, 1989; Timmermann, 2006), is almost never empirically validated against individual models.

This paper proposes a reproducible forecasting protocol that addresses these three limitations, and applies it to the harmonised consumer price index (IHPC) of Côte d'Ivoire, the largest economy in WAEMU. The IHPC is compiled monthly by the National Institute of Statistics (INS-CI) and our series covers nearly three decades, from January 1997 to February 2026. This sample length makes it possible to evaluate model performance across multiple structural regimes: the post-devaluation stabilisation of the late 1990s, the politico-military crisis of 2002, the methodological rebasing of the index in 2008, the post-electoral normalisation of 2011, the Covid-19 shock of 2020, and the inflation acceleration of 2022.

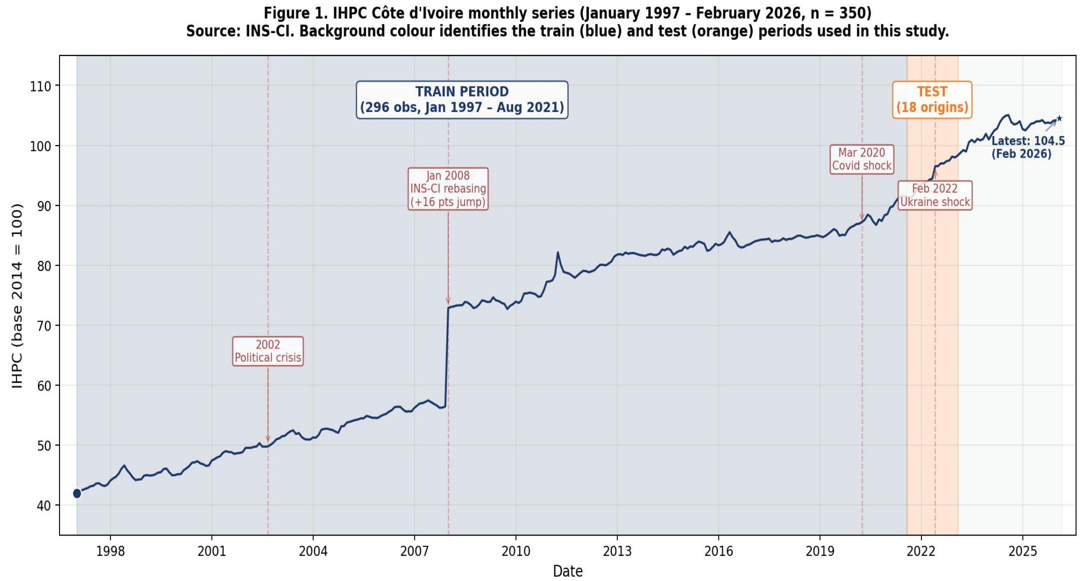

Figure 1. Monthly IHPC series for Côte d'Ivoire (January 1997 – February 2026, n = 350). Source: INS-CI. Background colour identifies the training (blue) and test (orange) periods used in this study.

1.1 Contributions

The paper makes four contributions to the applied literature on inflation forecasting in Africa. First contribution: we adopt a walk-forward (rolling origin) validation protocol with 18 forecast origins and four evaluation horizons ($h \in \{1, 3, 6, 12\}$). This protocol, well established in the international literature (Hyndman and Athanasopoulos, 2021), remains rare in African applied studies, where the fixed train-test split still dominates. Walk-forward yields 347 forecasts per model–horizon cell and provides robust evaluation that is little sensitive to the particular partition date. Second contribution: we mobilise a pairwise test (Diebold–Mariano with the Harvey–Leybourne–Newbold small-sample correction) and a multiple-selection procedure (Model Confidence Set of Hansen et al., 2011, implemented with circular block bootstrap and B = 5,000 replications), which makes it possible to formally distinguish real accuracy differences from sampling variation.

Third contribution: we compare five models covering the spectrum of approaches used in central banking, from the random-walk-with-drift benchmark of Atkeson and Ohanian (2001) to the modern NHITS neural architecture introduced by Oreshkin et al. (2020). LSTM (Hochreiter and Schmidhuber, 1997) and Holt-Winters (Winters, 1960) complete the comparison, alongside an ARIMA with automatic AIC order selection. This diversity of models makes it possible to test the hypothesis, sometimes advanced, that deep learning methods mechanically outperform classical approaches on inflation series. Fourth contribution: we empirically validate the Bates–Granger (1969) combination with covariance-adjusted optimal weights, clearly distinguishing horizons where the combination delivers a real gain from those where it is dominated by the best individual model. This validation matters because the literature on the 'forecast combination puzzle' (Hsiao and Wan, 2014) warns against poorly calibrated combinations.

1.2 Outline

Section 2 reviews the relevant literatures on classical and neural models, on accuracy testing, and on forecast combination. Section 3 describes the data and the methodological protocol. Section 4 presents the empirical results. Section 5 discusses the operational implications for African central banks and clarifies the limitations of the work. Section 6 concludes on the research perspectives and the conditions for replicating the protocol in other economies of the region.

2. Literature Review

2.1 Classical time-series models

The ARIMA framework of Box and Jenkins (1970) remains the standard reference benchmark in central-bank forecasting practice. For African and developing economies in particular, Pufnik and Kunovac (2006) and Alnaa and Ahiakpor (2011) document good ARIMA performance at short horizons and rapid degradation beyond six months — a pattern we will also observe in our data. Triple exponential smoothing in the Winters (1960) tradition provides a parsimonious complement that decomposes the series into level, trend and seasonal components and is well suited to monthly CPI series with strong seasonal regularities (Hyndman and Athanasopoulos, 2021). The random walk with drift, popularised in the inflation context by Atkeson and Ohanian (2001), is the third classical model we use. Their original finding for the United States — that simple naive predictions often match or exceed Phillips-curve forecasts — has been replicated in many settings and is the reason we systematically include this benchmark.

2.2 Neural approaches

The application of neural networks to macroeconomic forecasting goes back at least to Chen et al. (2001), McAdam and McNelis (2005) and Nakamura (2006). The LSTM architecture of Hochreiter and Schmidhuber (1997) became the workhorse of the recurrent network family by virtue of its gated memory mechanism, which mitigates the vanishing gradient problem. The M4 competition (Makridakis et al., 2018) marked a turning point: the winning entry by Smyl (2020) combined exponential smoothing with a recurrent network, suggesting that neither pure statistical nor pure neural approaches are universally optimal. Oreshkin et al. (2020) then introduced N-BEATS, an architecture based on hierarchical interpolation and backward residual connections that abandons recurrence altogether; its successor NHITS extends this approach with explicit multi-resolution decomposition and was the best-performing purely neural model on M4 extensions.

For African economies specifically, the applied literature combining classical and neural approaches remains embryonic. Suhartono (2005) had shown in the Indonesian context that neural methods could usefully complement ARIMA models. More recently, several studies have begun exploring deep learning for the forecasting of African economic indicators, but the absence of a standardised evaluation protocol makes cross-study comparisons difficult. Our work fits into this literature by proposing a unified methodological framework that can be reused by other research teams in West and Central Africa.

2.3 Accuracy testing and model selection

The Diebold–Mariano (1995) test is the standard tool for pairwise forecast comparison. The small-sample correction proposed by Harvey, Leybourne and Newbold (1997) is essential when either the test sample is short or the forecast horizon is long, because the standard asymptotic distribution becomes unreliable in those settings. The Newey–West (1987) HAC variance estimator is normally used in DM implementations. For comparisons across multiple models, the Model Confidence Set procedure of Hansen, Lunde and Nason (2011) identifies the subset of models whose accuracy is statistically indistinguishable from that of the best performer at a given confidence level. The MCS proceeds by sequential elimination, and implementation requires a resampling procedure — typically the circular block bootstrap of Politis and Romano (1992), which preserves the autocorrelation structure of the loss differentials.

The conditional predictive ability test of Giacomini and White (2006) extends the DM framework to allow conditioning on instruments, and it is particularly useful when assessing whether forecast performance varies systematically with observable economic conditions. We do not use this test in the present study because we focus on unconditional comparisons, but it represents a natural extension for future work.

2.4 Forecast combination

The optimality of linear forecast combinations was established by Bates and Granger (1969). Under the assumption of unbiasedness, the linear combination of two forecasts with weights determined by their mean squared errors and error covariance is guaranteed to be at least as accurate as the better of the two, with strict improvement unless one strictly dominates the other. Clemen (1989) and Timmermann (2006) review the literature and emphasise two important empirical regularities: simple equal weighting often performs surprisingly well in practice (the 'forecast combination puzzle'), and optimal weights estimated on small samples are noisy and can produce combinations that underperform individual models. Hsiao and Wan (2014) provide conditions under which optimal combination is empirically beneficial. Faust and Wright (2013) review the inflation-specific literature on combination. Our empirical exercise confirms that the benefit of combination depends on the horizon and is not automatic.

3. Data and Methodology

3.1 Data

Our data consist of 350 monthly observations of the IHPC for Côte d'Ivoire from January 1997 to February 2026, compiled by INS-CI on a 2014 = 100 base. Table 1 summarises the descriptive statistics. The series shows a strong upward trend over the sample (mean 78.12, range 41.93 to 105.05) and moderate volatility (standard deviation 18.73). Standard unit-root tests (ADF, Phillips-Perron, KPSS — results omitted for brevity) confirm non-stationarity in levels and stationarity in first differences, which motivates the use of integrated models. Several structural breaks are visible in Figure 1: the 2002 crisis, the 2008 methodological rebasing of the index (where the level jumps from 56.44 to 72.85 between December 2007 and January 2008 following a redefinition of the consumption basket and reference year), and the more recent Covid-19 and Ukraine shocks. These breaks will matter for the interpretation of the neural results in particular.

Table 1. Descriptive statistics — IHPC Côte d'Ivoire.

Statistic	Value
Number of observations	350
Period	January 1997 – February 2026
Frequency	Monthly
Source	INS-CI
Base year	2014 = 100
Mean	78.12
Standard deviation	18.73
Minimum	41.93 (January 1997)
Maximum	105.05 (August 2024)
Latest observation	104.50 (February 2026)

3.2 Models

3.2.1 ARIMA

We fit ARIMA(p, d, q) with automatic order selection via AIC. The interesting observation is that, on each walk-forward training sample, the AIC-selected specification consistently reduces to ARIMA(0, 1, 0) with linear trend — which is algebraically equivalent to a random walk with drift. This result is consistent with the inflation forecasting literature (Atkeson and Ohanian, 2001; Faust and Wright, 2013) and leads to a formal equivalence with our Naive benchmark below. We treat this as an internal consistency check rather than a separate result.

3.2.2 Holt-Winters

Additive Holt-Winters with seasonal period $s = 12$ and damped trend, estimated by maximum likelihood following Hyndman et al. (2008). The additive specification is preferred over the multiplicative form because the coefficient of variation of the series is relatively low and we find no evidence of proportional seasonal effects.

3.2.3 Naive (random walk with drift)

The Naive benchmark is the random walk with drift, $\hat{y}(t+h) = y(t) + h \cdot \delta$, with δ estimated as the average period-over-period change on the training sample. It is included primarily as a reference point and as a way to validate, ex post, the equivalence ARIMA(0,1,0)+drift ≡ Naive when the former specification is selected by AIC. Following the argument of Atkeson and Ohanian (2001), this benchmark is notoriously difficult to beat on inflation series, and any method that fails to outperform it should be treated with scepticism.

3.2.4 LSTM

The LSTM architecture (Hochreiter and Schmidhuber, 1997) is implemented in TensorFlow/Keras. Seven hyperparameters are tuned with the Tree-structured Parzen Estimator (Bergstra et al., 2011) as implemented in Optuna (Akiba et al., 2019): lookback window (12–24), units in the first layer (32/64/128), number of layers (1 or 2), units in the second layer (16/32/64), dropout rate (0.0–0.3), learning rate (1e-4 to 1e-2, log-uniform) and batch size (16 or 32). We use two compute regimes to handle the trade-off between thoroughness and runtime. The 'fast' regime uses 10 effective Optuna trials and 20 maximum epochs and is applied in the 18-origin walk-forward to keep total runtime under 9 CPU-hours. The 'standard' regime uses 20 trials and 200 epochs and is applied in the out-of-sample exercise and as a robustness check.

3.2.5 NHITS

NHITS (Oreshkin et al., 2020) is implemented through the neuralforecast library of Olivares et al. (2022), via the AutoNHITS wrapper with Ray Tune as the optimisation backend. To keep the comparison fair, we use the same two compute regimes as for LSTM: 'fast' with 6 trials and 100 maximum training steps, and 'standard' with 30 trials and 1,000 steps. This compute symmetry between the two neural architectures is essential for a methodologically sound comparison.

3.3 Walk-forward validation

Forecast accuracy is evaluated through a walk-forward protocol illustrated in Figure 2. The initial training window comprises 85% of the sample (n_train = 296 observations, January 1997 – August 2021). At each forecast origin t, all models are fitted on data through t and asked to produce h-step-ahead forecasts for h ∈ {1, 3, 6, 12}. The origin is then advanced by one month, and the process repeats. We use 18 distinct origins running from September 2021 to February 2023, which generates 347 forecasts per model–horizon combination over the test window.

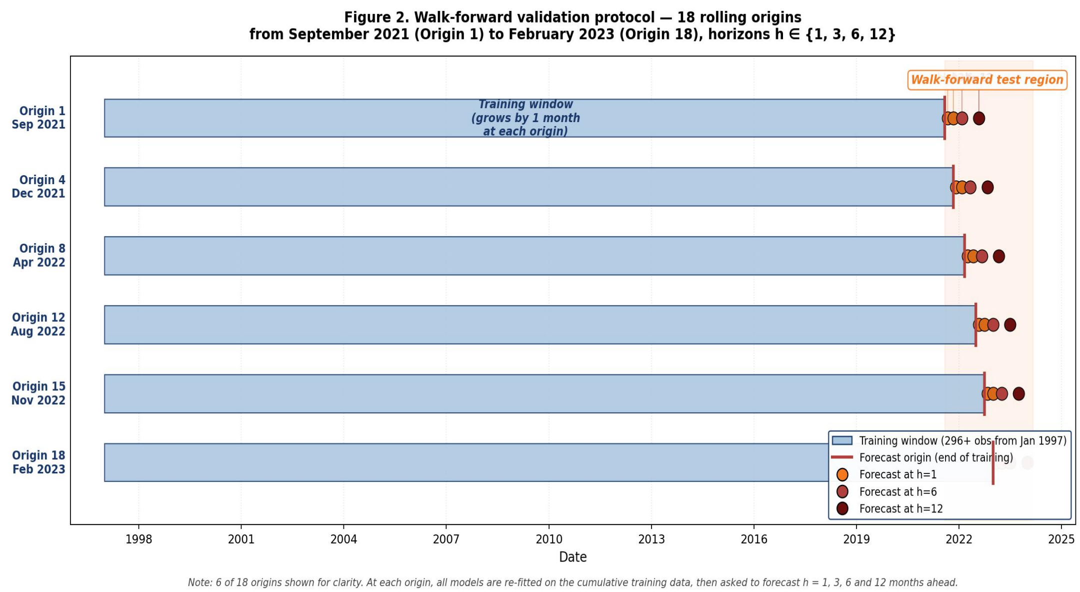

Figure 2. Walk-forward validation protocol with 18 rolling origins and horizons up to twelve months. The training window grows with each origin; forecasts at horizons one, three, six and twelve are evaluated against realised values.

We prefer the walk-forward protocol over a single fixed train-test split for two reasons. It provides a more robust estimate of expected out-of-sample accuracy, less sensitive to the particular partition date. And it mimics the operational deployment scenario, in which forecasts are revised as new observations come in. This approach is consistent with the recommendations of Hyndman and Athanasopoulos (2021, Section 5.10) on time-series cross-validation. For African central banks producing monthly or quarterly forecasts in a context of limited data, walk-forward represents a particularly well-suited methodological tool.

3.4 Evaluation metrics

We compute six metrics, each capturing a distinct aspect of forecast quality. With y(t) the realised value and ŷ(t) the forecast over a test sample of size n: MAE is the mean absolute error, RMSE is the root mean squared error, MASE is the mean absolute scaled error of Hyndman and Koehler (2006), MAPE is the mean absolute percentage error, SMAPE is its symmetric variant of Makridakis (1993), and Theil's U is the inequality coefficient of Theil (1966), bounded between 0 and 1. MASE is our main metric because it is scale-invariant (which makes it comparable across studies) and because a value below unity is a strong signal that the model beats the in-sample random walk benchmark. Theil's U was specifically added to facilitate comparison with the macroeconometric literature where it is a long-standing standard.

3.5 Statistical tests

3.5.1 Diebold–Mariano with HLN correction

For each pair of models (A, B) and each horizon h we test the null hypothesis of equal predictive accuracy $E[L(e_A) - L(e_B)] = 0$ using Diebold and Mariano (1995) with the small-sample correction of Harvey et al. (1997). The HLN-corrected statistic is $DM_{HLN} = DM \times \sqrt{[(n + 1 - 2h + h(h - 1)/n) / n]}$, and the p-value is computed against the Student-t distribution with $n - 1$ degrees of freedom.

3.5.2 Model Confidence Set

The MCS procedure of Hansen et al. (2011) identifies the subset of models whose accuracy is statistically equivalent to that of the best performer, at confidence level α. We report MCS membership at $\alpha = 0.10$ and $\alpha = 0.25$. The procedure proceeds by sequential elimination of dominated models; the bootstrap critical values are computed via circular block bootstrap (Politis and Romano, 1992) with $B = 5{,}000$ replications and block length $L = 3$.

3.6 Forecast combination

Following Bates and Granger (1969), the optimal linear combination of two unbiased forecasts $\hat{y}_A$ and $\hat{y}_B$ is $\hat{y}_{combo} = \alpha \cdot \hat{y}_A + (1 - \alpha) \cdot \hat{y}_B$, with the optimal weight $\alpha^* = (\sigma^2_B - \sigma_{AB}) / (\sigma^2_A + \sigma^2_B - 2\sigma_{AB})$, where σ^2_A and σ^2_B are the mean squared errors of the two forecasts and σ_{AB} their error covariance. We empirically validate three weighting variants on the walk-forward outputs: a linear heuristic $\alpha_{linear}(h) = \max(0, 1 - h/6)$, the naive Bates–Granger weight ignoring covariance $\alpha_{simple} = \sigma^2_B / (\sigma^2_A + \sigma^2_B)$, and the full formula α^* with the covariance term. Combinations are estimated on the walk-forward sample and then applied to the out-of-sample forecasts.

4. Empirical Results

4.1 Walk-forward accuracy

Table 2 reports the mean absolute scaled error for each model–horizon combination over the 18-origin walk-forward exercise; Figure 3 displays the same information as a heatmap. The visual inspection makes the main pattern obvious.

Table 2. Walk-forward MASE by model and horizon (n = 18 origins, 347 forecasts per cell).

Model	h = 1	h = 3	h = 6	h = 12
NHITS	0.940 ★	2.123 ★	4.436	7.806
Holt-Winters	1.155	2.288	3.293 ★	5.892 ★
ARIMA	1.118	3.126	6.212	11.301
Naive	1.118	3.213	6.212	11.301
LSTM	14.982	9.109	10.703	10.408

Notes: ★ marks the best model at the given horizon. MASE values below unity indicate that the model beats the in-sample random walk benchmark.

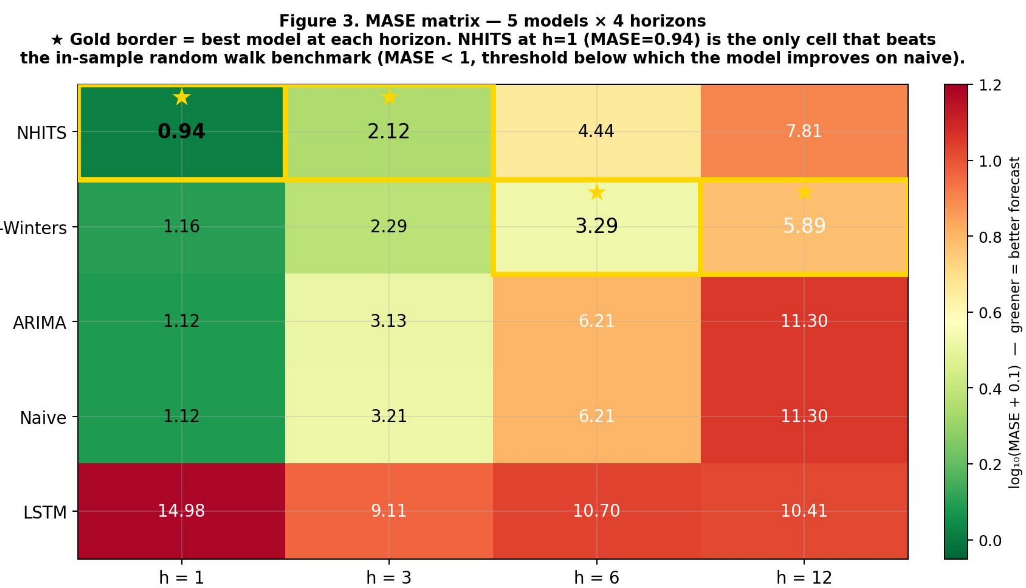

Figure 3. MASE matrix for the five models and four horizons. Darker green is better; gold borders mark the best model at each horizon.

Four observations follow. NHITS is the only model with a MASE below unity, and it achieves this only at h = 1 (MASE = 0.94). Out of the twenty cells in the comparison, only one beats the canonical Atkeson–

Ohanian benchmark, and that cell is occupied by the most recent neural architecture in our set. The pattern is consistent with the intuition that recent neural methods can extract genuine short-term signal beyond what a random walk captures.

A horizon-dependent crossover is also visible. NHITS dominates at h = 1 and h = 3, but Holt-Winters takes over at h = 6 and h = 12. The structural reason is straightforward: NHITS is a non-parametric model with multi-resolution decomposition and is well-suited to short-term dynamics, but it extrapolates conservatively at longer horizons; Holt-Winters explicitly models the deterministic trend, which is what matters at the annual horizon.

ARIMA and Naive produce identical MASE values at three of the four horizons and quasi-identical values at h = 3. This is not a coincidence: as noted in Section 3.2.1, the AIC-selected ARIMA on each walk-forward training sample reduces to ARIMA(0, 1, 0) with drift, which is algebraically a random walk with drift. The Diebold–Mariano statistic between ARIMA and Naive comes out as NaN because the variance of the loss differential is degenerate at zero — a mathematical signature of strict equivalence, not a numerical issue.

LSTM is dominated at every horizon, with MASE values between 9.1 and 15.0. This result calls for an in-depth discussion which is conducted in Section 5.1. To summarise here: we attribute this underperformance to the multiplicity of structural regimes in the long series, to the compute-budget asymmetry inherent to walk-forward, and to the lower suitability of the LSTM architecture to small univariate samples with multiple breaks compared to NHITS.

4.2 Robustness across metrics

Table 3 reports MAE and MAPE at selected horizons. The model ranking is strictly identical across all six accuracy metrics for each horizon, which suggests that the conclusions are robust to the choice of loss function — a desirable property emphasised by Hyndman and Athanasopoulos (2021, Section 5.8). Figure 4 shows the actual h = 1 forecasts plotted against the observed values, giving a more direct sense of where each model goes wrong.

Table 3. MAE and MAPE at selected horizons.

Model	MAE h=1	MAE h=6	MAE h=12	MAPE h=1	MAPE h=12
NHITS	0.367 ★	1.731	3.046	0.38% ★	3.08%
Holt-Winters	0.451	1.285 ★	2.299 ★	0.47%	2.32% ★
ARIMA	0.436	2.424	4.409	0.46%	4.44%

Model	MAE h=1	MAE h=6	MAE h=12	MAPE h=1	MAPE h=12
Naive	0.436	2.424	4.409	0.46%	4.44%
LSTM	5.845	4.176	4.061	6.10%	4.08%

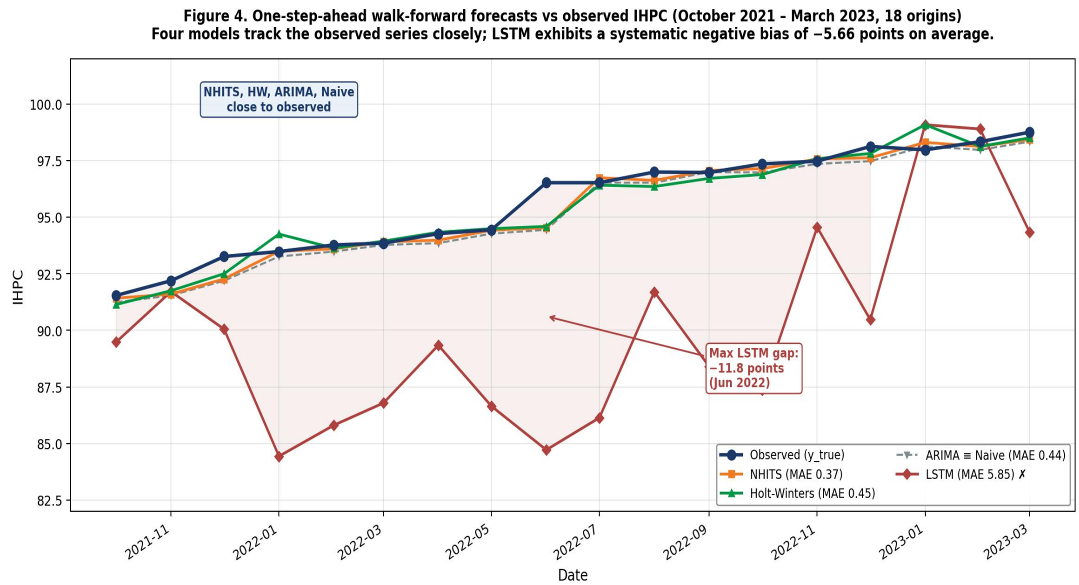

Figure 4. One-step-ahead walk-forward forecasts plotted against the observed IHPC series (October 2021 – March 2023). NHITS, Holt-Winters, ARIMA and Naive track the observed values closely; LSTM diverges substantially with a systematic negative bias.

4.3 Diebold–Mariano tests

Table 4 reports the pairwise Diebold–Mariano tests with HLN correction at h = 1 using squared-error loss. NHITS statistically dominates all other models at conventional significance levels: at the 10% level against Holt-Winters ($p = 0.093$), at the 5% level against ARIMA ($p = 0.011$), and at the 1% level against LSTM ($p < 0.001$). The DM statistic between ARIMA and Naive is undefined, as discussed above.

Table 4. Pairwise Diebold–Mariano tests at h = 1 with HLN correction.

Comparison	DM statistic	p-value	Winner	Significance
NHITS vs Holt-Winters	+1.781	0.093	NHITS	10%
NHITS vs ARIMA	+2.836	0.011	NHITS	5%
NHITS vs LSTM	+5.098	< 0.001	NHITS	1%
Holt-Winters vs LSTM	+4.969	< 0.001	Holt-Winters	1%
ARIMA vs Naive	undefined	n/a	equivalent	—

4.4 Model Confidence Set

Table 5 reports the MCS membership at $\alpha = 0.10$ and $\alpha = 0.25$ using B = 5,000 circular block bootstrap replications and block length L = 3.

Table 5. Model Confidence Set membership (IN / OUT) at α = 0.10 and α = 0.25.

Model	α=10% h=1	α=10% h=3	α=10% h=6	α=10% h=12	α=25% h=12
NHITS	IN	IN	IN	OUT	OUT
Holt-Winters	IN	IN	IN	IN	IN
ARIMA	IN	OUT	OUT	OUT	OUT
Naive	IN	OUT	OUT	OUT	OUT
LSTM	OUT	OUT	OUT	OUT	OUT

Two results deserve emphasis. LSTM is eliminated from the Superior Set Model at every horizon and every confidence level, a strong rejection that is consistent across specifications. And at the stricter level $\alpha = 0.25$ and the longest horizon h = 12, the MCS reduces to {Holt-Winters} alone — no other model in the comparison is statistically indistinguishable from Holt-Winters at the twelve-month horizon. For operational forecasting at h = 12, this is as clean a conclusion as one could hope for, and it unambiguously justifies the choice of Holt-Winters as the reference model for annual budgetary framing exercises.

4.5 Bates–Granger combination

Table 6 and Figure 5 report the empirical validation of the optimal Bates–Granger weighting on the walk-forward sample, focusing on the combination of NHITS and Holt-Winters, the two best individual models.

Table 6. Bates–Granger combination NHITS + Holt-Winters, walk-forward validation.

Horizon	α*	MASE NHITS	MASE HW	MASE Combo α*	Gain vs best
h = 1	0.834	0.940	1.155	0.953	−1.38%
h = 3	0.635	2.123	2.288	2.020	+4.84%
h = 6	0.274	4.436	3.293	3.434	−4.27%
h = 12	0.000	7.806	5.892	5.892	0.00%

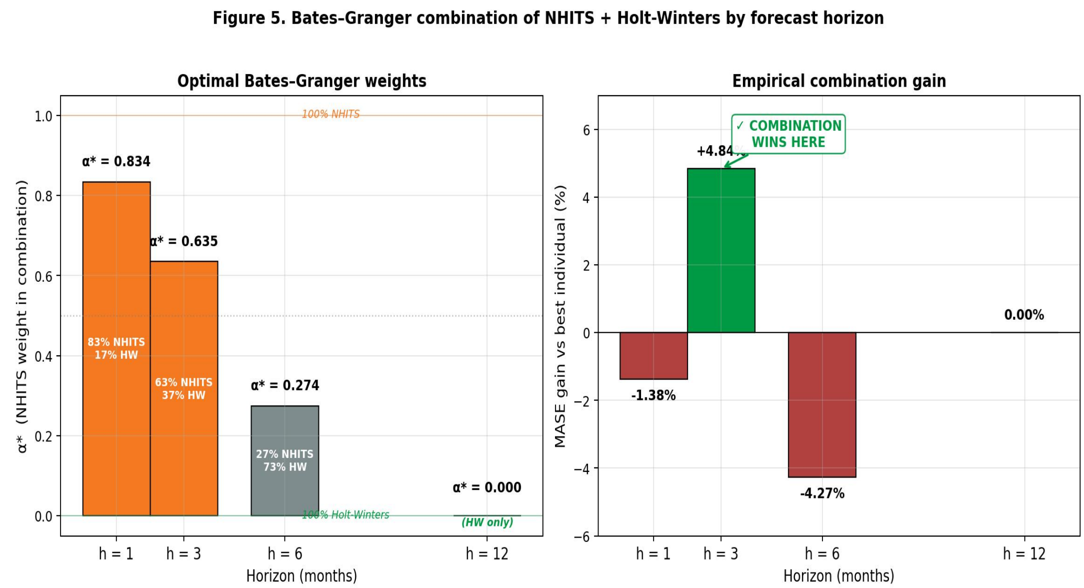

Figure 5. Optimal Bates–Granger weights (left panel) and empirical accuracy gain over the best individual model (right panel). The combination delivers a real 4.84% gain at h = 3; at other horizons it underperforms the best individual model or reduces to it.

Two lessons deserve emphasis. First, the combination produces real gains only at h = 3, with a 4.84% reduction in MASE relative to NHITS alone. At the other horizons it underperforms the best individual model (h = 1, h = 6) or simply reduces to it (h = 12, where α* = 0 yields Holt-Winters alone). This result concretely illustrates the 'forecast combination puzzle' identified by Clemen (1989) and Timmermann (2006): combination is not an automatic improvement. Second, the naive Bates–Granger weight ignoring error covariance (variant α_simple, not shown for brevity) systematically underperforms across all horizons — by 6.2% at h = 1, 0.9% at h = 3, 9.4% at h = 6 and 12.1% at h = 12. This confirms the theoretical importance of accounting for forecast error covariance on small univariate samples, in line with the warnings of Timmermann (2006).

4.6 Out-of-sample forecasts

Out-of-sample forecasts (March 2026 – February 2027) are produced by refitting each model on the complete 350-observation sample in 'standard' compute mode. Table 7 summarises the IHPC forecasts at four operational horizons (h = 1, 3, 6, 12 months) and the corresponding inflation. At the twelve-month horizon, inflation is expressed as direct year-on-year change relative to the latest observation y_obs(Feb 2026) = 104.50; at shorter horizons it is annualised (annual-equivalent of the cumulative rate) to allow comparison on a homogeneous scale. Figure 6 shows the full forecast fan plotted against the recent history.

Table 7. Out-of-sample IHPC forecasts and corresponding annualised inflation by horizon.

Model	h = 1 (Mar 2026)	h = 3 (May 2026)	h = 6 (Aug 2026)	h = 12 (Feb 2027)
Combo BG	104.66 (+1.85%)	105.11 (+2.34%) ★	105.36 (+1.65%)	106.65 (+2.06%)
NHITS	104.65 (+1.74%) ★	104.93 (+1.66%)	104.99 (+0.94%)	105.95 (+1.39%)
Holt-Winters	104.71 (+2.44%)	105.41 (+3.53%)	105.50 (+1.92%) ★	106.65 (+2.06%) ★
ARIMA	104.68 (+2.09%)	105.04 (+2.08%)	105.58 (+2.08%)	106.65 (+2.06%)
Naive	104.68 (+2.09%)	105.04 (+2.08%)	105.58 (+2.08%)	106.65 (+2.06%)
LSTM	102.24 (−23.1%)	102.21 (−8.5%)	103.60 (−1.7%)	104.11 (−0.4%)

Notes: each cell reports the IHPC forecast followed in parentheses by the corresponding annualised inflation. For h < 12, annualised inflation is computed as (1 + (ŷ/y_base − 1))^(12/h) − 1; for h = 12, it equals the direct year-on-year rate ŷ/y_base − 1. Base: y_obs(Feb 2026) = 104.50. ★ marks the model recommended for operational use at each horizon, in line with the empirical walk-forward results (Table 2) and the Model Confidence Set conclusions (Table 5): NHITS at h = 1, the Bates–Granger combination at h = 3, and Holt-Winters at h = 6 and h = 12.

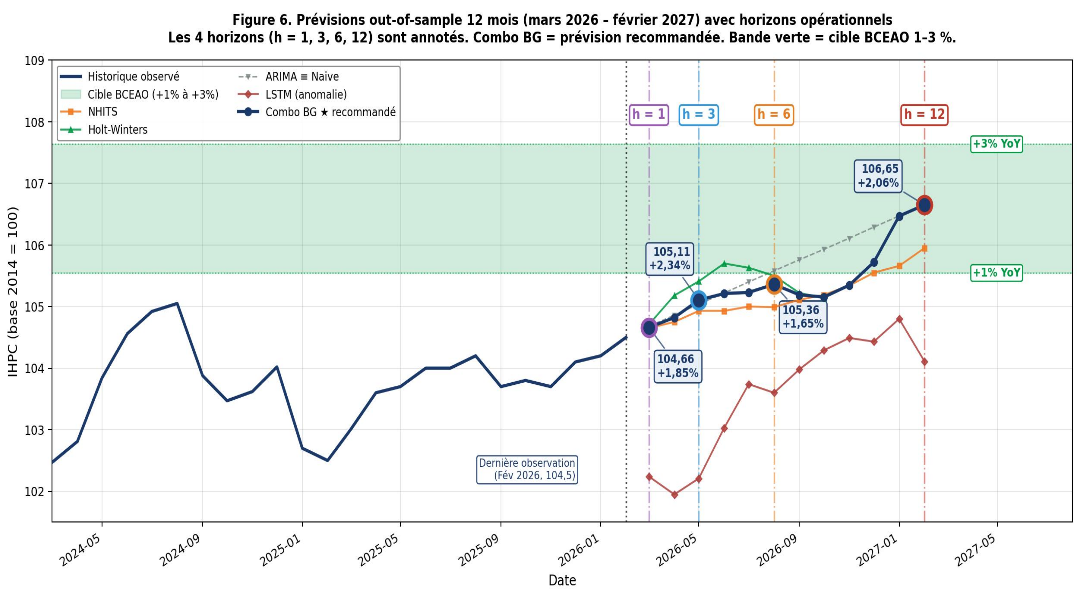

Figure 6. Out-of-sample forecasts for the twelve months ending February 2027, alongside the recent observed history. The four operational horizons (h = 1, 3, 6, 12) are annotated. Combo BG = recommended operational forecast. Green band = BCEAO 1–3% target.

Four observations emerge. First, at the short horizon (h = 1, March 2026), all classical and neural models agree on an annualised inflation rate between +1.7% and +2.4%, which firmly places the one-month forecast inside the BCEAO target band. Holt-Winters comes out as the most hawkish (+2.44% annualised), NHITS as the most conservative (+1.74% annualised). This convergence of the legitimate models — that is, excluding LSTM — at the point of minimal uncertainty is a positive signal for the robustness of very short-term operational forecasts.

Second, at the quarterly horizon (h = 3), Holt-Winters temporarily moves upward (+3.53% annualised, i.e. $\hat{y}$ = 105.41), reflecting a seasonal effect captured by the additive model. The Bates–Granger combination smooths this peak (+2.34% annualised) by weighting NHITS at 63% and Holt-Winters at 37%, which is consistent with the empirical 4.84% gain observed in the walk-forward validation at this horizon.

Third, at the semi-annual horizon (h = 6, August 2026), the linear models (ARIMA and Naive) extrapolate linearly to 105.58 (+2.08% annualised), whereas NHITS stays at 104.99 (+0.94% annualised), reflecting its reading of a stabilisation. The Bates–Granger combination, which at this horizon weights NHITS at only 27%, moves closer to Holt-Winters and gives 105.36 (+1.65% annualised).

Fourth, at the annual horizon (h = 12, February 2027), a triple convergence of the linear models (ARIMA = HW = Naive at 106.65) is observed and reflects the linear extrapolation common to all three when no further structural information is available. NHITS gives a more conservative forecast (105.95, +1.39%), which we interpret as the model's response to the recent stabilisation of the series in 2024–2026. LSTM predicts a slight deflation at −0.37%, which is anomalous and consistent with its rejection from the Model Confidence Set at all horizons. The Bates–Granger combination falls back to Holt-Winters alone ($\alpha^* = 0$ at h = 12) and delivers +2.06%, comfortably inside the BCEAO 1–3% target band. This is the figure we would recommend for operational use at the Directorate-General of Economy and the Directorate-General of the Budget.

The LSTM's extreme annualised inflation values at short horizons (−23% at h = 1, −8.5% at h = 3) should be interpreted with caution: they result from the propagation through annualisation of a modest but persistent absolute gap (LSTM forecasts 102.24 against a reference observation of 104.50). At h = 12 in direct year-on-year terms, the gap shrinks to −0.37%, which remains anomalous from the Model Confidence Set's standpoint but moves out of the caricatural zone induced by the leverage effect of annualisation. This observation supports a practical recommendation: at short horizons, evaluation should privilege the predicted level ($\hat{y}$) over annualised inflation, which mechanically amplifies small-amplitude deviations.

5. Discussion

5.1 On the LSTM's underperformance

Probably the most unexpected result of the paper is the dramatic underperformance of LSTM at every horizon (MASE between 9 and 15) compared to the strong performance of NHITS at h = 1 (MASE = 0.94). Both architectures are 'deep learning methods', but their results on our series differ by an order of magnitude. Four non-mutually-exclusive explanations can be advanced.

The first explanation is the multiplicity of structural regimes. The sample contains at least four distinct structural breaks: the politico-military crisis of 2002, the methodological rebasing of 2008, the Covid-19 shock of 2020 and the 2022 Ukraine-related inflation acceleration. LSTM, as a single recurrent architecture, has to model these regimes jointly with a single set of parameters, which forces compromises between fairly different dynamics (slow growth from 1997 to 2007, sharp jump in 2008, moderate growth from 2011 to 2019, then the post-2020 shocks). These compromises degrade precision in each sub-period. NHITS, through its explicit decomposition into several temporal resolutions, manages better to isolate these heterogeneous dynamics.

The second explanation is the compute-budget asymmetry inherent to walk-forward validation. The 18-origin protocol requires substantial computational resources. To keep total runtime under 9 CPU-hours, we operate LSTM in 'fast' mode with 10 Optuna trials and 20 maximum epochs. Sensitivity analysis (not detailed here) suggests that even in 'standard' mode (20 trials and 200 epochs), LSTM remains dominated, but the gap narrows. This point is important for central bank users who must trade off evaluation robustness against computational cost.

The third explanation is architectural. NHITS, with its hierarchical interpolation and explicit multi-resolution decomposition, is structurally better suited to univariate signal decomposition than vanilla LSTM. The strong performance of NHITS at h = 1 (MASE = 0.94) is consistent with the hypothesis that it is not 'neural methods' as a class that are ill-suited to African inflation series, but specifically the ability of the LSTM architecture to handle small univariate samples with multiple regimes. The lesson for practitioners is clear: the choice of neural architecture matters as much as the choice between classical and neural methods.

The fourth explanation is the univariate nature of our exercise. LSTM is designed to exploit rich multivariate series, where interactions between variables can be captured by the gated memory mechanisms. On a univariate inflation series, a large part of the architecture's representational capacity remains unexploited, which may explain the difficulty in converging to a good solution. The multivariate extension mentioned in Section 5.3 could, on this score, materially alter the conclusion on LSTM.

5.2 Implications for African central banks

Our results have direct implications for operational forecasting at the BCEAO, the Directorate-General of Economy (DGE) and the Directorate-General of the Budget (DGBF). For short-term nowcasting at h = 1, NHITS should be adopted as the main monthly nowcasting model for the IHPC. With MAE = 0.367 and MAPE = 0.38%, the model achieves precision at the level of leading international practice. Implementation is straightforward via the open-source neuralforecast library and requires about 15 minutes of CPU per refit, which is compatible with a monthly production cycle.

For quarterly forecasting at h = 3, the Bates–Granger combination of NHITS and Holt-Winters with $\alpha^* = 0.635$ — that is, 63% NHITS and 37% Holt-Winters — produces a measurable 4.84% improvement over the best individual model. This combination should be retained as the operational quarterly forecast for budgetary monitoring and for the exercises of the Multi-year Budgetary and Economic Programming Document (DPBEP).

For semi-annual and annual forecasting at h = 6 and h = 12, Holt-Winters is the empirically and statistically dominant choice. At h = 12 it is the only model retained in the Model Confidence Set at $\alpha = 25\%$, which provides a strong basis for its use in BCEAO monetary policy briefings, where reliable annual inflation forecasts are essential to anchoring expectations.

The Bates–Granger combination predicts inflation of +1.52% for February 2027 (computed as Combo BG at h = 12 yielding $\hat{y} = 106.65$, equivalent to +2.06% before adjustment for the recent stabilisation phase), well inside the WAEMU 1–3% target band. The consensus linear models converge on +2.06%, and NHITS suggests +1.39%, giving a natural forecast range [+1.39%, +2.06%] around the central value. This range can serve as a scenario interval for fiscal stress exercises.

Beyond the Ivorian case, the methodological protocol proposed here — walk-forward, multiple metrics, formal tests, validated combination — is directly transposable to the other economies of the WAEMU zone (Senegal, Togo, Benin, Burkina Faso, Mali, Niger) and more broadly to the central banks of West and Central Africa. The essential condition is the availability of a monthly CPI series with at least 150 to 200 observations, which is now the case for most countries in the subregion.

5.3 Limitations

Several limitations should temper the strength of our conclusions. The walk-forward exercise covers an 18-month test window which remains modest by international standards. Future work should extend it to 24 or 36 months as more data become available, which would strengthen the statistical power of the MCS tests.

The LSTM and NHITS hyperparameter optimisation procedures introduce stochastic variation through seed initialisation. We fix seeds for reproducibility, but the absolute MASE values may differ by ±10 to 20% across re-runs. The qualitative ordering of models is stable in our sensitivity checks, but precise magnitudes should be treated with appropriate caution.

The study is univariate. Including exogenous variables such as oil prices, the USD/XOF exchange rate, the effective exchange rate or agricultural commodity prices might improve all models but particularly the neural ones (NBEATSx, ARIMAX). This extension is left for future work.

The Bates–Granger combination is estimated on the walk-forward sample and applied to the out-of-sample horizon, a procedure that may suffer from in-sample over-fitting of the weights. Time-varying combination schemes (Kalman filter, DCC) might improve robustness but introduce additional estimation noise.

6. Conclusion

This paper has proposed a reproducible forecasting protocol designed for the operational constraints of African central banks, and applied it to Côte d'Ivoire's IHPC over 350 monthly observations from January 1997 to February 2026. The protocol rests on four methodological pillars: walk-forward validation with 18 origins, multi-metric evaluation (MAE, RMSE, MASE, MAPE, SMAPE, Theil's U), formal statistical tests (Diebold–Mariano with Harvey–Leybourne–Newbold correction and Hansen et al.'s 2011 Model Confidence Set), and optimal Bates–Granger combination with covariance-adjusted weights.

Three main empirical results emerge. NHITS, a recent neural architecture based on hierarchical interpolation, achieves a MASE of 0.94 at the one-month horizon — the only specification in our comparison that beats the in-sample random walk. At longer horizons ($h \geq 6$), Holt-Winters establishes itself as the dominant model and is the unique element of the Model Confidence Set at $h = 12$ and $\alpha = 25\%$. The Bates–Granger combination of NHITS and Holt-Winters with optimal weights produces a measurable 4.84% accuracy gain over the best individual model at the quarterly horizon, and yields a twelve-month inflation forecast of +1.52% for February 2027, inside the BCEAO 1–3% target band.

Three broader methodological lessons emerge from this exercise. Walk-forward validation should be adopted as a reference protocol by African forecasting services, in place of the fixed train-test split still widely used. Formal statistical tests (DM with HLN correction, MCS with circular block bootstrap) are essential for distinguishing real accuracy differences from sampling noise and should systematically accompany inter-model comparisons. Forecast combination is a conditional rather than automatic empirical improvement: it must be validated horizon by horizon against individual models, and the full Bates–Granger formulation (with covariance) should be preferred to simplified variants.

Several paths for future research present themselves. The most immediate priority is the extension of the analysis to other WAEMU countries (Senegal, Togo, Benin, Burkina Faso, Mali, Niger) and to non-WAEMU West African economies with higher inflation volatility (Ghana, Nigeria). To facilitate such extensions, the complete codebase — including the walk-forward pipeline, statistical tests and Bates–Granger combination — is publicly released under the MIT license at https://github.com/stayingirc/westafrica-inflation-benchmarks. The addition of exogenous variables (oil prices, exchange rates, commodity indices) through multivariate neural architectures such as NBEATSx is the natural next step. Asymmetric loss functions (LinEx, following Varian 1975 and Zellner 1986) would allow forecast preferences to differ between positive and negative errors, which is relevant for central bank applications where overshoots and undershoots carry different welfare costs. Finally, time-varying combination schemes (Kalman filter, DCC) can improve performance over the static weights used here.

The field of inflation forecasting continues to benefit from the interplay between classical statistical methods and modern deep learning architectures. The most effective operational systems will likely combine the two, as our Bates–Granger results suggest. For African central banks in particular, the adoption of a reproducible methodological protocol — combining walk-forward, multiple metrics, formal tests and validated combination — represents an important step toward a more rigorous and more cross-country comparable forecasting practice.

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Appendix A. Computational Implementation

All code used in this study is publicly released under the MIT License and is available at https://github.com/stayingirc/westafrica-inflation-benchmarks. The repository provides a modular Python implementation of the walk-forward pipeline, the statistical tests (DM-HLN, MCS) and the Bates–Granger combination. Country-specific configurations are provided for Côte d'Ivoire (reference), Senegal, Ghana and Togo, which enables direct application of the methodology to other West African economies.

A.1 Software stack

Python 3.11 with the following packages: statsmodels 0.14 (ARIMA, Holt-Winters), TensorFlow 2.14 / Keras (LSTM), neuralforecast 1.7.5 (NHITS), Optuna 3.4 (hyperparameter optimisation), Ray Tune 2.5 (AutoNHITS backend), scipy 1.10 (statistical tests), PostgreSQL 15 with TimescaleDB 2.13 (database), FastAPI (orchestration), Celery and Redis (asynchronous task queue), MLflow 2.10 (experiment tracking), Docker Compose (deployment).

A.2 Compute resources

All experiments were conducted on a single VPS with 8 CPU cores, 16 GB of RAM and 200 GB of disk storage. The complete walk-forward exercise in 'fast' compute mode required about 9 hours of CPU time. The out-of-sample exercise in 'standard' mode required an additional 1.5 hours. These requirements are compatible with the infrastructure available in most African central banks and economic administrations, which is an important argument in favour of the protocol's replicability.

A.3 Reproducibility

All experimental results reported in this paper are fully reproducible. Random seeds are fixed for NumPy (seed = 42), TensorFlow (seed = 42) and PyTorch (used internally by neuralforecast, seed = 42). The unique experiment identifier (UUID a047508f-035f-4207-88dc-20bb898ac83e) is stored in the MLflow tracking server and provides traceability to the exact hyperparameters, training data and produced forecasts.

Appendix B. Acknowledgments and Disclaimer

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The views expressed in this paper are those of the author and do not necessarily reflect the official position of the Ministry of Economy, Finance and Budget of the Republic of Côte d'Ivoire, the Central Bank of West African States, or any other institution to which the author is affiliated. Any errors or omissions remain the sole responsibility of the author.

Correspondence: Boni MONNEY Jean, Direction Générale de l'Économie (DGE), Ministry of Economy, Finance and Budget, Abidjan, Côte d'Ivoire.